

SRI KRISHNA KALYAN BANDARU

Data Analyst | Adelaide, SA, Australia | Full Working Rights (485 Visa)

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PROFESSIONAL SUMMARY

Data Analyst with a Master's degree in Artificial Intelligence & Machine Learning (University of Adelaide) and a track record of turning complex, messy datasets into clear, decision-ready insights. Skilled in SQL, Python and Power BI/Tableau, with hands-on experience diagnosing why business performance shifts, pressure-testing experiment design before results are acted on, and building dashboards the business actually uses. Comfortable working across operational, technical and business teams to gather requirements and deliver measurable outcomes, backed by an engineering background and two years keeping live production systems running.

SKILLS

Data Analysis: SQL (BigQuery, MySQL), Python (pandas, NumPy), Power BI, Tableau, Excel (pivot tables, what-if modelling), Statistical Analysis, A/B Testing

Process Improvement: Root Cause Analysis, Incident Triage & Escalation, Gap Analysis, Continuous Improvement, SAP (production support)

Reporting & Insights: Data Visualisation (Matplotlib, Seaborn), Dashboard Development, Performance Reporting

Stakeholder Engagement: Requirements Gathering, Cross-Functional Collaboration, Presenting Complex Findings to Non-Technical Audiences

Technical: Python, SQL, Machine Learning, Azure, AWS, Agile/Scrum, CI/CD, ROS2

PROFESSIONAL EXPERIENCE

Data & Systems Analyst, Research Placement

Feb 2025 – Aug 2025

Auto-ID Lab, University of Adelaide | Adelaide, Australia

- Ported the open-source EGO-Planner trajectory-planning stack (5 core ROS packages) from ROS1 to ROS2 and re-targeted the controller layer from PX4 to ArduPilot firmware, unblocking UAV research on Ubuntu 22.04 / JetPack 6.2 where the original ROS1 stack could no longer build
- Built 3 Gazebo warehouse environments for pre-flight validation; after repeated simulated failures, isolated the cause as feature-tracking collapse in the low-texture environment and replaced the default VINS-Fusion tracker with a SuperPoint neural tracker, lifting tracked features per frame from about 5 to 2,000 against the 15–20 minimum needed for stable position estimation
- Analysed planner performance across simulated runs to document 4 systematic failure modes and benchmarked 3 candidate planning algorithms against them, informing the team's algorithm selection for subsequent development
- Trained and benchmarked 5 object-detection configurations on the VisDrone dataset (548 validation images, 28,000+ instances), quantifying the accuracy/compute trade-off for edge deployment: a MobileNetV2 backbone cut inference cost 76% (21.2 to 5.04 GFLOPs) at half the mAP50, informing model selection for onboard hardware

Machine Learning Engineer Intern, Data & Model Performance Analysis

Aug 2024 – Nov 2024

Proto Corp Pvt Ltd | Remote, India

- Assembled and annotated a 3,508-image construction-defect dataset across 5 defect classes using polygon segmentation in Roboflow, exporting to both YOLO YAML and COCO formats so a single labelled corpus fed both model families without re-annotation
- Trained and benchmarked 4 instance-segmentation architectures on a 519-image / 1,514-instance validation set, selecting YOLOv8m-seg on an accuracy-vs-compute basis: 2.1x the mask mAP50-95 of Mask R-CNN (0.259 vs 0.122) at roughly half the compute (110 vs 200–300 GFLOPs), at 10.5 ms per image on a Tesla T4
- Diagnosed that three YOLO variants clustered within 0.007 mask mAP50-95 despite a 43% spread in compute, concluding that further preprocessing and fine-tuning had hit an architectural rather than a tuning ceiling; recommended Vision Transformers, which the team adopted as its next direction
- Built a FastAPI backend (3 endpoints) with a Streamlit frontend, self-hosted on team machines, letting testers and developers without model access run any of the 3 trained models and export defect-boundary coordinates as JSON for 3D overlay; handed over trained weights and serving code that the team carried into the production build

Project Engineer, SAP Production Support

May 2018 – Sept 2019

Wipro Technologies | India

- Monitored SAP transaction queues on a 24/7 rotating roster, triaging several hundred processing errors per shift and diagnosing each failure well enough to route it to the responsible team (ABAP, CLICK, Plant Maintenance) or to onsite stakeholders in the UK
- Tracked recurring and high-volume error patterns filling the queue and escalated them with supporting detail, so fixes addressed the underlying defect rather than repeatedly clearing the same failure
- Took over coordination of a 5 person support team as senior colleagues rotated off the project, owning the shift roster, acting as escalation point to management for incidents, and producing end-of-shift reports on error volumes, resolutions and open escalations for handover

DATA ANALYTICS PROJECTS

Business Analytics, Olist Marketplace

SQL, Power BI, Statistics

- With review scores crashing for two months and sellers wrongly assumed to be at fault, built a 13-query SQL request bank across five stakeholder teams and traced the collapse to a carrier delivery failure concentrated in Rio de Janeiro
- Quantified the business impact: late orders scored 1.7 stars lower on average, around 2,500 late orders and R\$350K of GMV affected, with 54% of late orders receiving a one-star review, giving stakeholders an evidence-backed case for where to act
- Reviewed an A/B test read-out, identified that the experiment's randomisation was broken before any conclusions were drawn, and delivered the corrected findings through a 3-page Power BI dashboard and an Excel what-if model

E-Commerce Business Intelligence, Brazil Market

SQL, Google BigQuery

- Analysed 100,000 e-commerce orders across 8 relational datasets (customers, orders, payments, freight, geolocation) using SQL on Google BigQuery to identify a 137% increase in order value from 2017 to 2018 as a key business performance indicator
- Wrote complex multi-table JOIN queries, window functions, and date-based aggregations to surface order seasonality trends (peak months: May, July, August), regional freight cost disparities across 27 Brazilian states, and delivery performance gaps
- Translated findings into specific, actionable business recommendations, presented through structured outputs built for a commercial audience

Content Strategy Analysis, Netflix Catalogue

Python, Exploratory Data Analysis

- Ran end-to-end exploratory analysis on 8,800+ Netflix titles using Python (pandas, Matplotlib, Seaborn) to surface content distribution trends, seasonal addition patterns, and genre performance
- Facing extensive missing data, designed a multi-layered groupby-mode imputation strategy that preserved data integrity across a 202,000-row unnested dataset
- Delivered data-backed recommendations on optimal release timing (December and July, aligned with holiday peaks), genre prioritisation, and content frequency, communicated in plain business language for an executive audience

Predictive Analytics, Alzheimer's Disease Diagnosis

Python, Statistical Modelling

- Built a predictive analytics pipeline achieving 81% accuracy in identifying Alzheimer's progression from complex, ambiguous neurological signal data
- Applied rigorous statistical validation (leave-one-patient-out cross-validation, AUC 0.74) to ensure findings were reliable before drawing conclusions in a high-stakes clinical domain

EDUCATION

Master of Artificial Intelligence & Machine Learning

Dec 2025

University of Adelaide | GPA: 6.1/7 (Distinction average) | Focus: Data Analytics, Machine Learning, NLP, Statistical Modelling

Master of Science in Mechatronics Engineering

2022

University of Rome 'Tor Vergata' | Grade: 106/110 | Focus: Systems Analysis, Process Automation, Control Systems

Bachelor of Mechanical Engineering

2018

KL University, India | First Class with Distinction | Focus: Engineering Systems, Data Structures, Problem Solving